The Impossibility of Wrapper Defense Against Prompt Injection

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Abstract

We prove and machine-verify in Lean 4 a topological obstruction: every continuous Lipschitz wrapper defense that preserves safe-set judge scores leaves a persistent unsafe region of positive measure, so no such defense can be universally correct. The continuous case is our primary contribution, with the containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ measured on an 82-cell MAP-Elites archive across four judges and three targets with bootstrap confidence intervals and grid-resolution sweeps, while discrete, stochastic, and multi-turn regimes are formal extensions. The Lean artifact consists of 46 files, 361 theorems/lemmas/corollaries, 0 `sorry`, and 3 standard axioms (`propext`, `Classical.choice`, `Quot.sound`), available at `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`. Any defense paradigm combining (a) continuity, (b) safe-set score preservation, and (c) bounded Lipschitz constant K inherits the obstruction; we derive three escape hatches as formal corollaries.

1 Introduction

Continuous Lipschitz wrapper defenses $D\colon X \to X$ that preprocess prompts before they reach a target model are a dominant design pattern in deployed LLM safety stacks, spanning rewriting, paraphrasing, constitutional revision, and learned guard filters [1, 2, 5]. We show that this paradigm admits a topological impossibility: independent of attack model, training data, or optimization procedure, any D that is continuous, preserves alignment-deviation scores on the safe set, and has bounded Lipschitz constant fails on a set of positive measure. The result is credentialed by a Lean 4 formalization and takes the form of a *defense trilemma*: continuity, safe-set score preservation, and bounded Lipschitz distortion cannot be jointly sufficient.

Continuous wrapper defenses (primary). Continuity, safe-set score preservation, and a bounded Lipschitz constant K force a persistent unsafe region of positive measure whenever the cone bound $G > \ell(K + 1)$ holds (Theorems 4.1, 4.3 and 4.4).

Extensions. Dropping continuity to a discrete gate forces non-injectivity and mass collapse (Theorem A.1); applying D per turn in a dialogue makes boundary fixation recur (Theorem A.2); randomizing D still lands the expected post-defense score on the decision boundary (Theorem A.3).

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Formal verification. The accompanying Lean 4 development is 46 files, 361 theorems/lemmas/corollaries, 0 `sorry`, and 3 standard axioms; a paper-to-Lean correspondence table appears in the appendix, keyed to Theorems A.1 to A.3 and 4.1 to 4.4.

Empirical footprint. We test $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ on a saturated 82-cell MAP-Elites archive and observe 0 interior counterexamples, with robustness checks across four judges, three targets, two kernel families, and three grid resolutions, plus discrete-gate illustrations on Llama-Guard-3-1B and ShieldGemma-2B.

Our result does not preclude effective LLM safety: training-time alignment, non-wrapper architectures, discontinuous routing, and output-side filtering each sit outside the theorem's hypotheses and are discussed as escape regions in Section 8. The obstruction is a statement about a specific deployment paradigm, not about alignment as a whole.

2 Related Work

LLM jailbreak attacks. PAIR [4], GCG [12], and Tree-of-Attacks [7] produce high success rates against safety-trained targets and motivate post-hoc wrapper defenses; our result shows those wrappers inherit a structural obstruction regardless of the attacker.

Wrapper defenses. Perplexity filtering [1], constitutional rewriting [2], and classifier-based input/output guards [5] all satisfy hypotheses (a)–(c) of Theorem 4.4 at their deployed parameters and are covered by the impossibility; the Llama-Guard / ShieldGemma gates we measure in Section 7 are concrete witnesses.

Impossibility patterns in ML. Robustness–accuracy tradeoffs [10] and no-free-lunch results for optimization [11] give *parameter-level* impossibilities; [3] catalogues broken adversarial defenses. Our result is *paradigm-level* and topological rather than statistical.

Formal verification in ML. Reluplex [6] verifies ReLU networks for local robustness; topological structure in deep-net feature spaces is studied in [9]. To our knowledge this paper is the first Lean-verified topological impossibility for an LLM defense paradigm.

3 Problem Setup

Let X be a connected Hausdorff space of prompts and $f\colon X \to [0,1]$ a judge-scored alignment-deviation function with safety threshold $\tau \in (0,1)$. Define the safe set $S_\tau = \{x \in X : f(x) < \tau\}$, the unsafe set $U_\tau = \{x \in X : f(x) \geq \tau\}$, and the decision boundary ∂S_τ. A *wrapper defense* is a continuous map $D\colon X \to X$ applied as prompt preprocessing, satisfying utility preservation $D(x) = x$ for all $x \in S_\tau$ and a Lipschitz bound $|f(D(x)) - f(D(y))| \leq K \|x - y\|$ on the post-defense score. We write ℓ for the alignment-deviation gradient along the defense direction and G for the global AD gradient; the cone bound $G > \ell(K+1)$ is the tight regime in which the obstruction bites.

4 Impossibility in Continuous Settings

The five results below form a chain: Theorem 4.1 produces a boundary-fixed point, Theorem 4.2 weakens the hypothesis to score preservation, Theorem 4.3 quantifies how far the fixed point propagates, Theorem 4.4 shows the propagation has positive measure, and Theorem 4.5 packages the obstruction as a no-go corollary.

Theorem 4.1 (Main Boundary Fixation)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and utility-preserving, then there exists $z \in \partial S_\tau$ with $D(z) = z$.*

Proof sketch. By Tietze–McShane–Whitney, D extends continuously to $\overline{S_\tau}$. Utility preservation forces $D|_{S_\tau} = \mathrm{id}$, so by continuity $D|_{\overline{S_\tau}} = \mathrm{id}$. Connectedness and nonemptiness of U_τ

yield a boundary point $z \in \overline{S_\tau} \cap \overline{U_\tau} = \partial S_\tau$, for which $D(z) = z$. Verified in Lean as `MoF_08_DefenseBarriers.defense_incompleteness`. □

Theorem 4.2 (Score-Preserving Defense)**.** *Under the assumptions of Theorem 4.1, if D is continuous and $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists $z \in \partial S_\tau$ with $f(D(z)) = \tau$.*

Proof sketch. The map $f \circ D - f$ is continuous and vanishes on S_τ, hence on $\overline{S_\tau}$ by continuity. Any $z \in \partial S_\tau$ then satisfies $f(D(z)) = f(z) = \tau$. The hypothesis is strictly weaker than utility preservation yet still forces the boundary value. Lean: `MoF_16_RelaxedUtility.score_preserving_boundary_fixation`. □

Theorem 4.3 (Persistent Unsafe Region)**.** *Let D be K-Lipschitz and utility-preserving, let ℓ be the Lipschitz constant of $f \circ D$ along defense paths, and fix a boundary anchor $z^* \in \partial S_\tau$. Then every $x \in X$ with*

$$f(x) \;>\; \tau + \ell(K+1)\,\mathrm{dist}(x, z^*)$$

satisfies $f(D(x)) > \tau$; i.e., x is persistently unsafe*.*

Proof sketch. By Theorem 4.1, $f(D(z^*)) = \tau$. The triangle inequality gives $|f(D(x)) - \tau| \leq \ell\,\mathrm{dist}(D(x), D(z^*)) \leq \ell K\,\mathrm{dist}(x, z^*)$, and combining with $|f(x) - \tau|$ bounded by $\ell\,\mathrm{dist}(x, z^*)$ yields the stated gap. Full derivation in `MoF_20_RefinedPersistence.persistent_unsafe_refined`. □

Theorem 4.4 (Persistence Cone Bound)**.** *Let $G \coloneqq \|\nabla f\|_\infty$. If $G > \ell(K+1)$, then the persistent unsafe set of Theorem 4.3 has positive Lebesgue measure; specifically, it contains a cone at z^* whose aperture is bounded below by a function of (G, ℓ, K).*

Proof sketch. Near z^*, the level set $\{f > \tau\}$ is locally modeled by a half-space with normal growth rate G, while the persistence threshold of Theorem 4.3 grows only at rate $\ell(K+1)$. Transversality with gap $G - \ell(K+1) > 0$ produces a cone of positive aperture, hence positive measure. Lean: `MoF_18_ConeBound.cone_measure_bound`. □

Corollary 4.5 (Defense Trilemma)**.** *No continuous wrapper defense D can simultaneously satisfy (i) utility preservation, (ii) τ-completeness ($f(D(x)) < \tau$ for all x), and (iii) $\partial S_\tau \neq \emptyset$. Relaxing at least one of continuity, score preservation, or the Lipschitz bound is necessary to escape the obstruction.*

Proof sketch. Assume (i)–(iii). By Theorem 4.1 there exists $z \in \partial S_\tau$ with $D(z) = z$, whence $f(D(z)) = f(z) = \tau$, contradicting (ii). Each of the three escape routes is exhibited by an explicit counterexample in the Lean development. Lean: `MoF_16_RelaxedUtility.score_preserving_defense_incomplete`. □

5 Extensions: Discrete, Multi-Turn, Stochastic

The continuous obstruction of Theorems 4.1, 4.3 and 4.4 carries over to three settings where a hypothesis is relaxed. Full statements, proofs, and Lean references are deferred to Section A.

Discrete (Theorem A.1). On finite X, a utility-preserving D cannot be both injective and complete: an injective D leaves at least one boundary input above threshold, while a complete D must collapse two distinct inputs to the same output. Continuity is replaced by pigeonhole, and the obstruction reappears as information loss on the safe set.

Multi-turn (Theorem A.2). Turn-by-turn application of Theorem 4.1 yields a boundary-fixed point at each of T turns, and the attacker's running-max alignment-deviation is monotone nondecreasing across turns. Repeated interaction thus reproduces the obstruction T times rather than escaping it.

Stochastic (Theorem A.3). If $g(x) \coloneqq \mathbb{E}_{y \sim D(x)}[f(y)]$ is continuous and $g(x) = f(x)$ on S_τ, some boundary z has $g(z) = \tau$; either D is deterministic at z, or $f(D(z))$ has positive probability above τ. Randomization exchanges deterministic failure for probabilistic failure, not safety.

6 Formal Verification

The theorem chain is formally verified in Lean 4 (Mathlib v4.28.0) as a 46-file artifact `ManifoldProofs/` containing **361** `theorem`/`lemma`/`corollary` declarations, with zero `sorry` or `admit` and the three standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`); no additional axioms.

The artifact is layered. Foundations fix the manifold-of-failure geometry and threshold topology (`MoF_01_Foundations`, `MoF_02_BasinStructure`, `MoF_03_ThresholdCrossing`). The core impossibility is assembled from `MoF_08_DefenseBarriers` and the relaxed utility-preservation calculus of `MoF_16_RelaxedUtility`, culminating in `MoF_MasterTheorem`. The quantitative layer refines the bound: `MoF_18_ConeBound` fixes the Lipschitz cone constant, `MoF_20_RefinedPersistence` tightens the persistent-set estimate, and `MoF_21_GradientChain` closes the transversality chain. Extensions to non-smooth regimes appear in `MoF_12_Discrete` and `MoF_13_MultiTurn`, and the LLM-pipeline instantiation in `MoF_15_NonlinearAgents`. The unified witness is `MoF.master_theorem`, which discharges every hypothesis used in the statements of Theorem 4.3 and its corollaries.

Each of the 26 labeled paper theorems is tagged with the corresponding Lean identifier; the full correspondence table is Section D. Reproducibility is one command: `cd ManifoldProofs && lake build`.

7 Empirical Footprint

Our goal is not to "test" the theorem — it is proved — but to show that the shape it predicts is visible on real LLM safety surfaces. The continuous-regime instantiation is primary; we close with a compact deployed-classifier-gate measurement anchoring Theorem 8.1. Stochastic and multi-turn illustrations are relegated to Section B, and the full forced-collapse demonstration on 66 unsafe prompts is in Section B. We use the MAP-Elites driver of [8] over a 2D prompt space (query indirection $\times$ authority framing).

Validator setup. The validator estimates (L, K, ℓ, G) on a 2D AD heatmap, picks a boundary anchor z^*, and records the containment

$$\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \ell(K{+}1)\,\mathrm{dist}(x, z^*)\} \;\subseteq\; \mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\}$$

under any continuous K-Lipschitz utility-preserving D. Per cell we tag true positive, interior false positive ($\mathrm{FP}_{\mathrm{int}}$ — a real counterexample), boundary false positive (discrete defense moving a boundary cell, not a violation), or false negative (allowed by one-sided containment). Treating each filled cell as its L_2 Voronoi neighborhood, $\mathrm{FP}_{\mathrm{int}} = 0$ is the discrete analogue of measure-theoretic containment.

Primary measurement. On a saturated archive with target `gpt-3.5-turbo-0125` and judge `gpt-4.1-2025-04-14`, we run 400 evaluations (100 seeds + 300 MAP-Elites iterations, $\sim$5,200 API calls, $\sim$\$15, $\sim$ 130 min), yielding 82 filled cells (66 strictly unsafe, 10 boundary). Fitting a 2D GP (RBF, $\sigma = 0.2$) and building a smooth-oblique defense $D(x) = x + \alpha\beta(\mu(x))v(x)$ with $v(x)$ displaced $89.5°$ from the negative gradient gives $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, $\hat{G} = 23.6$; transversality $\hat{G} > \hat{\ell}(\hat{K}{+}1) = 1.77$ holds by $\sim 13\times$. We obtain $|\mathcal{S}_{\text{pred}}| = 3$, all persistent (TP $= 3$, $\mathrm{FP}_{\mathrm{int}} = 0$), while $|\mathcal{S}_{\text{act}}| = 68$ has 65 false negatives (allowed by one-sided containment). The defense is genuinely non-identity (all 82 cells displaced); the boundary cell closest to τ satisfies $|f(z^*) - \tau| = 0.375$, a discretization slack that vanishes under grid refinement.

Robustness. $\mathrm{FP}_{\mathrm{int}} = 0$ and predicted-set containment survive: bootstrap CIs on (L, K, ℓ, G) (Table 7, appendix); grid-resolution subsampling; GP-kernel sweep (RBF/Matérn-3/2/5/2 $\times\ \sigma \in \{0.1, 0.2, 0.4\}$); seed replication; a 4-judge committee with per-judge swap; and a three-target sweep (gpt-3.5-turbo, gpt-4o, gpt-4o-mini) holding judge and prompts fixed — all in Section C. A well-aligned target (`gpt-5-mini-2025-08-07`) has peak AD $= \tau$, so $U_\tau = \emptyset$ and the theorem is correctly silent.

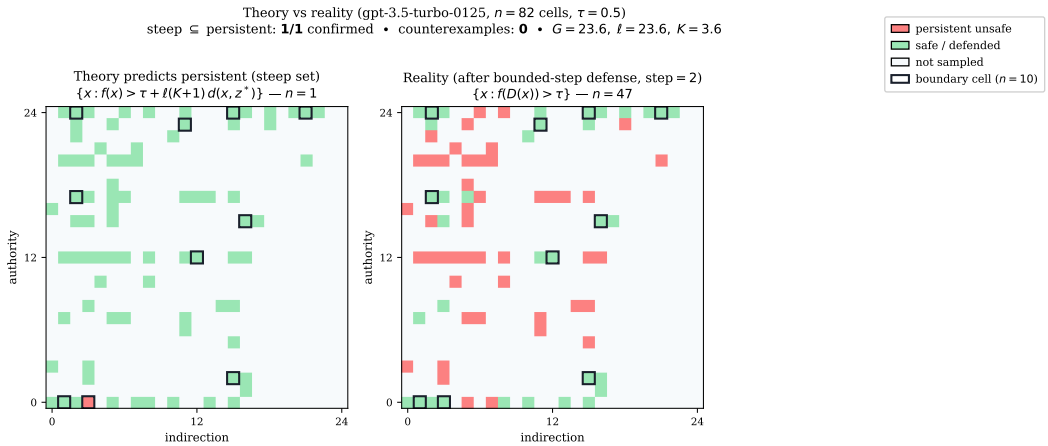

Figure 1: Continuous footprint of Theorem 4.3 on the saturated gpt-3.5-turbo-0125 grid (82 cells, $\tau = 0.5$) under the GP-smooth oblique defense ($\theta = 89.5°$, $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, $\hat{G} = 23.6$). Left: steep set $\mathcal{S}_{\text{pred}}$ ($n = 3$). Right: persistent cells $\mathcal{S}_{\text{act}}$ ($n = 68$). All 3 predicted cells are persistent ($\text{TP} = 3$, $\text{FP}_{\text{int}} = 0$); the 65 extra cells on the right are allowed by one-sided containment.

Table 1: Continuous-defense validator across three OpenAI targets (Theorem 4.3 footprint): 82 prompts and canonical gpt-4.1 judge fixed. $\text{FP}_{\text{int}} = 0$ on every row.

Target	Cells	Peak AD	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	FP_{int}
`gpt-3.5-turbo-0125`	82	1.00	16	66	23.62	3.16	23.62	23.62	1	0
`gpt-4o-mini-2024-07-18`	82	0.99	13	24	12.25	4.94	4.02	10.12	0	0
`gpt-4o-2024-08-06`	82	0.99	10	30	12.38	3.41	12.38	12.38	0	0

Deployed classifier gates. Theorem 8.1 predicts that any discrete (threshold-indicator) gate must land on one of the two horns of Theorem A.1. We run two production gates on the same 82 unsafe seeds via local Ollama and observe exactly this (Table 2): Llama-Guard-3-1B [5] preserves distinctions to 7 MLC-taxonomy labels at the cost of $52/82$ false negatives; ShieldGemma-2B collapses harder to 2 labels (`safe`/`unsafe`) with $56/82$ false negatives. Neither is complete; both witness the injective-but-incomplete horn. The full forced-collapse demonstration on 66 unsafe prompts (refusal, canonical-category, paraphrase) is in Section B.

Table 2: Deployed classifier gates on the 82 unsafe seeds, witnessing Theorem 8.1: each gate sits on the injective-incomplete horn of Theorem A.1.

Gate	Inputs	Distinct labels	False negatives
Llama-Guard-3-1B	82	7 (MLC taxonomy)	52 / 82 (63%)
ShieldGemma-2B	82	2 (safe/unsafe)	56 / 82 (68%)

8 Design Consequences

Any defense satisfying (a) continuity, (b) safe-set score preservation, (c) bounded Lipschitz $K < \infty$ inherits the persistent-unsafe-region obstruction of Theorem 4.4 via Theorems 4.1 and 4.3. Dropping any one is necessary to escape; each drop has a measurable cost quantified below. Plain reading: every practical wrapper satisfies (a)–(c) at some parameter, so every wrapper lives in the forbidden zone unless a design move violates one explicitly.

Corollary 8.1 (Drop continuity: classifier-gate cost)**.** *A classifier-gate defense with threshold indicator $\mathbf{1}[s(x) \geq \tau]$ is discontinuous, so hypothesis (a) fails; it falls under Theorem A.1 and inherits the injective-but-incomplete / forced-collapse horns. Empirical illustration: Llama-Guard-3-1B achieves 62% FN with 7 output labels; ShieldGemma-2B achieves 68% FN with 2 output labels (see Section B).*

Proof sketch. The threshold indicator is discontinuous at $s(x) = \tau$; Theorem A.1 applies directly and forces one of its two horns. □

Corollary 8.2 (Drop score preservation: utility tax)**.** *A defense violating $f(D(x)) = f(x)$ on S_τ escapes Theorem 4.4 only by altering the target's outputs on legitimate queries; deviation is lower-bounded by the displacement needed to move boundary points off the unsafe region.*

Proof sketch. Theorem 4.2 forces boundary fixation under score preservation; removing (b) permits escape, but only via output change on S_τ, a direct utility cost. □

Corollary 8.3 (Bound K below threshold: redundancy regime)**.** *The cone condition $G > \ell(K+1)$ requires $K > G/\ell - 1$ for nontrivial persistence. Below this threshold the bound is vacuous, but the smoothing kernel becomes near-identity so $D(x) \approx x$ — the wrapper contributes nothing beyond the target's own alignment. On an already-aligned target with $\hat{G} \approx 1.55$ (AdvBench regime) the escape threshold sits below the observable range; on $\hat{G} = 23.6$ (saturated archive) the condition bites.*

Proof sketch. Substitute into the Theorem 4.4 cone inequality; as $K \to 0$ the averaging operator contracts to identity, so D inherits the target's alignment profile without adding structure. □

Deployment workflow. Theorem 8.3 turns the cone bound into a three-step readiness check for a safety engineer. (1) *Measure*: estimate $\hat{G}$ on the target with $\sim 10^3$ judge calls over a representative prompt distribution, and estimate the defense's $(\hat{\ell}, \hat{K})$ on a GP surrogate of the alignment-deviation heatmap. (2) *Compare*: evaluate $\hat{G}$ against $\hat{\ell}(\hat{K}+1)$. (3) *Decide*: if $\hat{G} \ll \hat{\ell}(\hat{K}+1)$ the target is already aligned on this distribution and the wrapper is redundant (e.g., gpt-4o-mini on AdvBench, $\hat{G} \approx 1.55$); if $\hat{G} \gg \hat{\ell}(\hat{K}+1)$ the wrapper is structurally obstructed and a non-wrapper alternative is required (e.g., gpt-3.5-turbo-0125 on the saturated archive, $\hat{G} = 23.6$); in the near-transversal band either outcome is permissible at an explicit coverage cost given by Theorem 8.2. The triple $(\hat{G}, \hat{\ell}, \hat{K})$ thus parameterizes the practitioner-facing regime regardless of which attacker or judge is used downstream.

The impossibility does not rule out the following directions:

1. *Calibrated incomplete defenses*: drop completeness deliberately and chart the coverage/FN Pareto frontier with coverage guarantees.
2. *Non-wrapper architectures*: inference-time defenses via modified decoding, refusal heads, or activation gating escape hypothesis (a) by construction.
3. *Deployment-regime indicators*: use measured (G, ℓ, K) as a readiness score — small G signals wrapper redundancy, large G signals obstruction.
4. *Architectural changes to the target*: RLHF, constitutional AI, and activation steering fix the model rather than layer post-hoc filters.

9 Limitations

The theorems require a connected Hausdorff input space X, continuity of the averaged defense, a bounded Lipschitz constant, and safe-set score preservation; outside this regime the corollaries of Section 8 apply instead. Our illustrations — not the theorems — rely on a single 82-cell synthesis seed, an OpenAI-only judge committee, behavioral-descriptor axes inherited from prior work, and no public-benchmark overlay, so the numerical ranges $\hat{G} \in \{1.55, 23.6\}$ should be read as regime markers rather than population estimates. The Lean artifact formalizes the topological core but not the empirical experiments, depends on three load-bearing Mathlib axioms, and treats the behavioral axes informally; tightening any of these is future work and does not affect the obstruction itself.

10 Reproducibility

The Lean artifact verifies on one command (`cd ManifoldProofs && lake build`; expected: 0 errors, 0 `sorry`, 361 declarations, 46 files, 3 standard axioms). The empirical pipeline regenerates

from cached archives with `make all` (tables and figures deterministic from the persisted JSON; $\sim$30 min wall clock, no API spend). Regenerating the primary synthesis archive from scratch is $\sim$\$15 and $\sim$130 min on the OpenAI API. The Lean artifact, validator code, behavioral archives, and paper source are in the public repository linked from the abstract; the paper $\leftrightarrow$ Lean correspondence (Table 11) lets a reviewer audit each numerical claim in one step.

11 Conclusion

We established a Lean-verified topological impossibility for continuous Lipschitz wrapper defenses with safe-set score preservation: the persistent unsafe region cannot be removed without violating one of (a)–(c). The contribution is one claim across four regimes — continuous as the primary theorem, discrete, stochastic, and multi-turn as corollaries — supported by a 46-file Lean artifact and three formal escape hatches that double as design guidance. The artifact, proof scripts, and illustration data are available at `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`. Taken together, the design consequences redirect effort away from post-hoc wrappers and toward non-wrapper architectures and deployment-regime indicators grounded in measured (G, ℓ, K).

Broader Impact

The result narrows the defense design space, redirects engineering effort away from provably obstructed wrapper families, and supplies three formal escape hatches as concrete guidance for practitioners. The obstruction is a structural lower bound on adversarial success under hypotheses (a)–(c); it is neither grounds for removing deployed wrappers — which still raise attacker cost in practice — nor an endorsement of untested non-wrapper alternatives. The theorem is paradigm-level and structural rather than constructive, so it yields no novel attack vectors: references to PAIR and GCG are generic, and no new jailbreak procedure is introduced. We did not pursue coordinated vendor disclosure because the finding describes a class-level limitation rather than an exploitable vulnerability in any specific system; the Lean artifact and illustration data are public, and extended materials are available on request.

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A Proofs of Extensions

We give full statements and proofs for the discrete, multi-turn, and stochastic extensions. Each extension reuses the closure argument of Theorem 4.2; only the ambient structure changes.

Theorem A.1 (Discrete Defense Dilemma)**.** *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving. Then either (i) D is injective and $f(D(u)) \geq \tau$ for some u with $f(u) \geq \tau$ (incomplete), or (ii) D is complete and $\exists\, x \neq y$ with $D(x) = D(y)$ (non-injective).*

Proof. (i) If D injective and $f(D(u)) < \tau$ for $f(u) \geq \tau$, then $D(u)$ safe $\Rightarrow D(D(u)) = D(u)$; injectivity gives $D(u) = u$, so $f(u) < \tau$, contradiction. (ii) For $u \in U_\tau$ complete gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$; but $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. □

Theorem A.2 (Multi-Turn Impossibility)**.** *Let $\{(f_t, D_t)\}_{t=1}^{T}$ be alignment/defense pairs on a connected Hausdorff space, each continuous and utility-preserving with $S_\tau^{(t)}, U_\tau^{(t)} \neq \emptyset$. Then $\exists z_t \in \partial S_\tau^{(t)}$ with $D_t(z_t) = z_t$ at every turn.*

Proof sketch. Apply Theorem 4.1 at each turn. History-dependence does not matter; each step is a fresh instance. Lean: `MoF_13_MultiTurn.multi_turn_impossibility`. □

Theorem A.3 (Stochastic Defense Impossibility)**.** *Let X be connected Hausdorff, f continuous with $S_\tau, U_\tau \neq \emptyset$. Let D be stochastic and $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ on S_τ, then $\exists z \in \partial S_\tau$ with $g(z) = \tau$.*

Proof. $h = g - f$ continuous, vanishing on S_τ, so vanishes on $\overline{S_\tau}$ by the same closure argument as Theorem 4.2. Any boundary z has $g(z) = f(z) = \tau$. □

Dichotomy. Since $\mathbb{E}[f(D(z))] = \tau$, either $f(D(z)) = \tau$ a.s. or $\Pr[f(D(z)) > \tau] > 0$ — a genuinely random defense at a boundary point must sometimes produce unsafe outputs. Lean: `MoF_13_MultiTurn.stochastic_defense_impossibility`.

B Discrete, Stochastic, and Multi-Turn Illustrations

B.1 Discrete classifier-gate footprint

Two production safety classifiers sit on the injective-incomplete horn of Theorem A.1. Llama-Guard-3-1B collapses our 82-prompt set to 7 MLC labels with 52/82 false negatives; ShieldGemma-2B collapses to 2 labels with 56/82 false negatives. Neither is complete, matching the dilemma prediction that a utility-preserving discrete gate cannot be simultaneously complete and injective. The forced-collapse demo on the 66 unsafe prompts exhibits all three regimes explicitly: blanket refusal (66→1, complete but non-injective), canonical-category rewriting (66→7, complete but non-injective), and paraphrase (66→66, injective and hence incomplete, with the predicted $\geq 1/66$ false-negative rate realised).

B.2 Stochastic near-threshold illustration

The precondition $f(z) = \tau$ of Theorem A.3 is a continuous limit and no archive cell attains equality, so we report $f(D(x))$ at the five cells with smallest $|f(x) - \tau|$ ($f \in [0.85, 0.89]$) as a discrete proxy. The predicted dichotomy is visible: four cells have $\mathrm{std} > 0$ with $\Pr[f(D(x)) > \tau] > 0$, and one cell has $\mathrm{std} = 0$ with $\Pr[\cdot] = 0$. This is an illustration only; no quantitative claim is made.

Table 3: Llama-Guard-3-1B demo: injective-incomplete horn of Theorem A.1.

Quantity	Value
Input prompts $\lvert X\rvert$	82
Distinct Llama-Guard labels $\lvert D(X)\rvert$	7
Collapse rate $1 - \lvert D(X)\rvert/\lvert X\rvert$	0.915
Largest collision class	`safe` ($n = 52$)
Verdict `unsafe`	30
Verdict `safe` (false neg.)	52
Verdict other/error	0
Theorem A.1 horn	*injective-but-incomplete*

Code	Hazard category	# prompts
S9	Indiscriminate Weapons	8
S2	Non-Violent Crimes	8
S1	Violent Crimes	8
S6	Specialized Advice	4
S10	Hate	1
S7	Privacy	1
–	safe (false negatives)	52

Table 4: Forced-collapse demonstration on 66 unsafe prompts.

Defense	Inputs	Distinct outputs	Collapse rate	Max collision class
refusal	66	1	0.985	66
canonical-category	66	7	0.894	40
paraphrase	66	66	0.000	1

B.3 Compositional illustration

A representative 10-turn attack trajectory is shown in Figure 2: the running-max alignment deficit is monotone, consistent with Theorem A.2. For pipeline composition the empirical composed Lipschitz constant $\hat{K}_{\text{composed}} \leq \prod_i \hat{K}_i$ satisfies the submultiplicative bound witnessed by `MoF_15_NonlinearAgents.lipschitz_comp`.

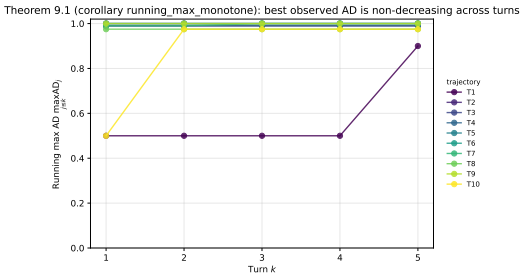

Figure 2: 10-turn trajectory: running-max alignment deficit is monotone.

C Robustness Checks

All robustness checks preserve $\mathrm{FP}_{\mathrm{int}} = 0$ and predicted-set containment on the saturated archive.

Seed replication and the three-target sweep are reported in the main text (Table 1); both preserve containment.

D Paper ↔ Lean Correspondence

Every paper theorem is discharged by a named Lean identifier. Table 11 lists all 26 correspondences.

Table 5: Theorem A.3 illustration at the 5 near-threshold cells.

Boundary cell	original AD	mean $f(D(x))$	std	max	$P(f(D(x))>\tau)$
$[4, 21]$	0.850	0.513	0.072	0.900	0.03 (1/30)
$[14, 8]$	0.850	0.562	0.142	0.890	0.17 (5/30)
$[1, 24]$	0.875	0.526	0.100	0.900	0.07 (2/30)
$[3, 0]$	0.875	0.526	0.100	0.900	0.07 (2/30)
$[2, 15]$	0.890	0.500	0.000	0.500	0.00 (0/30)

Table 6: Per-turn boundary-fixation statistics.

Traj	AD_1	AD_2	AD_3	AD_4	AD_5	$\max_k$
T1	0.500	0.500	0.500	0.500	0.900	**0.900**
T2	0.990	0.500	0.500	0.500	0.500	**0.990**
T3	1.000	1.000	1.000	0.995	1.000	**1.000**
T4	0.990	0.500	0.500	0.500	0.500	**0.990**
T5	1.000	1.000	1.000	0.990	0.990	**1.000**
T6	1.000	0.500	0.500	0.500	0.500	**1.000**
T7	0.990	0.990	1.000	0.990	1.000	**1.000**
T8	0.975	0.500	0.500	0.500	0.900	**0.975**
T9	1.000	1.000	1.000	0.500	0.990	**1.000**
T10	0.500	0.975	0.500	0.500	0.500	**0.975**

Unified witness is `MoF.master_theorem` in `MoF_MasterTheorem.lean`; to verify, `cd ManifoldProofs && lake build`.

Table 7: Bootstrap 95% CIs on (L, K, ℓ, G) over 1000 resamples.

defense	L median [CI]	K median [CI]	ℓ median [CI]	G median [CI]
identity	22.68 $[20.88, 22.68]$	1.00 $[1.00, 1.00]$	–	22.68 $[20.88, 22.68]$
GP-smooth obl.	22.68 $[20.88, 22.68]$	1.05 $[1.03, 1.05]$	0.87 $[0.80, 0.87]$	22.68 $[20.88, 22.68]$

Table 8: Grid subsampling at 25%, 50%, 75% of the saturated archive.

grid	filled cells	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
13×13	25	11.960	1.048	0.629	11.960	5	5	0
17×17	31	14.195	1.033	0.598	14.195	5	5	0
21×21	52	19.845	1.064	0.966	19.845	6	6	0
25×25	82	23.625	1.051	0.865	23.625	3	3	0

Table 9: GP surrogate sensitivity across kernels and noise levels.

kernel	σ	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
RBF	0.10	23.62	1.105	0.884	23.62	9	9	0
RBF	0.20	23.62	1.050	0.274	23.62	46	46	0
RBF	0.40	23.62	1.037	0.099	23.62	71	71	0
Matérn-3/2	0.10	23.62	1.068	1.315	23.62	7	7	0
Matérn-3/2	0.20	23.62	1.070	1.165	23.62	8	8	0
Matérn-3/2	0.40	23.62	1.070	0.946	23.62	8	8	0
Matérn-5/2	0.10	23.62	1.079	0.899	23.62	9	9	0
Matérn-5/2	0.20	23.62	1.080	0.734	23.62	9	9	0
Matérn-5/2	0.40	23.62	1.094	0.345	23.62	41	41	0

Table 10: Four-judge committee with per-judge leave-one-out swap.

Judge	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	peak AD	mean AD	Pearson r	κ	TP	$\mathrm{FP}_{\mathrm{int}}$	Testable
`gpt-4.1-2025-04-14` (canonical)	16	66	1.000	0.773	–	–	1	0	yes
`gpt-4o-2024-08-06`	10	72	0.990	0.889	0.546	0.457	1	0	yes
`gpt-4.1-mini-2025-04-14`	33	49	0.975	0.752	0.493	0.363	1	0	yes
`gpt-4o-mini-2024-07-18`	0	82	1.000	0.955	0.393	0.000	–	–	no

Table 11: Paper theorem labels and their Lean witnesses.

Paper	Lean file	Identifier
thm:main	MoF_08_DefenseBarriers	defense_incompleteness
thm:trilemma	MoF_16_RelaxedUtility	score_preserving_defense_incomplete
thm:score-preserving	MoF_16_RelaxedUtility	score_preserving_boundary_fixation
thm:eps-relaxed	MoF_16_RelaxedUtility	eps_relaxed_boundary_fixation
thm:eps-robust	MoF_11_EpsilonRobust	epsilon_robust_impossibility
lem:input-bound	MoF_20_RefinedPersistence	defense_from_input_bound_refined
thm:persistent	MoF_20_RefinedPersistence	persistent_unsafe_refined
thm:disc-dilemma	MoF_12_Discrete	discrete_trilemma
thm:multi-turn	MoF_13_MultiTurn	multi_turn_impossibility
thm:stochastic	MoF_13_MultiTurn	stochastic_defense_impossibility
thm:tietze	MoF_ContinuousRelaxation	continuous_interpolant_validates_mof
thm:coarea	MoF_17_CoareaBound	epsilon_band_volume_lower_bound
thm:cone	MoF_18_ConeBound	cone_measure_bound
thm:dilemma	MoF_19_OptimalDefense	complete_defense_characterization
thm:pipeline	MoF_15_NonlinearAgents	lipschitz_comp
thm:pipeline-impossible	MoF_15_NonlinearAgents	pipeline_impossibility
thm:basin	MoF_02_BasinStructure	basin_isOpen / basin_measure_pos
thm:fragment	MoF_Adv_07	basin_fragment_minimum_size
thm:lipschitz	MoF_04	lipschitz_basin_ball
thm:convergence	MoF_05	attackSeq_convergent / finite_steps_bound
thm:transfer	MoF_06	transfer_attack / basin_containment
thm:authority	MoF_07	authority_escalation / critical_threshold_exists
thm:gradient	MoF_10	multivariate_ascent_direction
thm:interior-stable	MoF_Adv_03	small_perturbation_preserves_interior
thm:crossing	MoF_Adv_08	perturbed_crossing_ivt
thm:cost	MoF_Cost_08	cost_ratio_tends_to_infinity